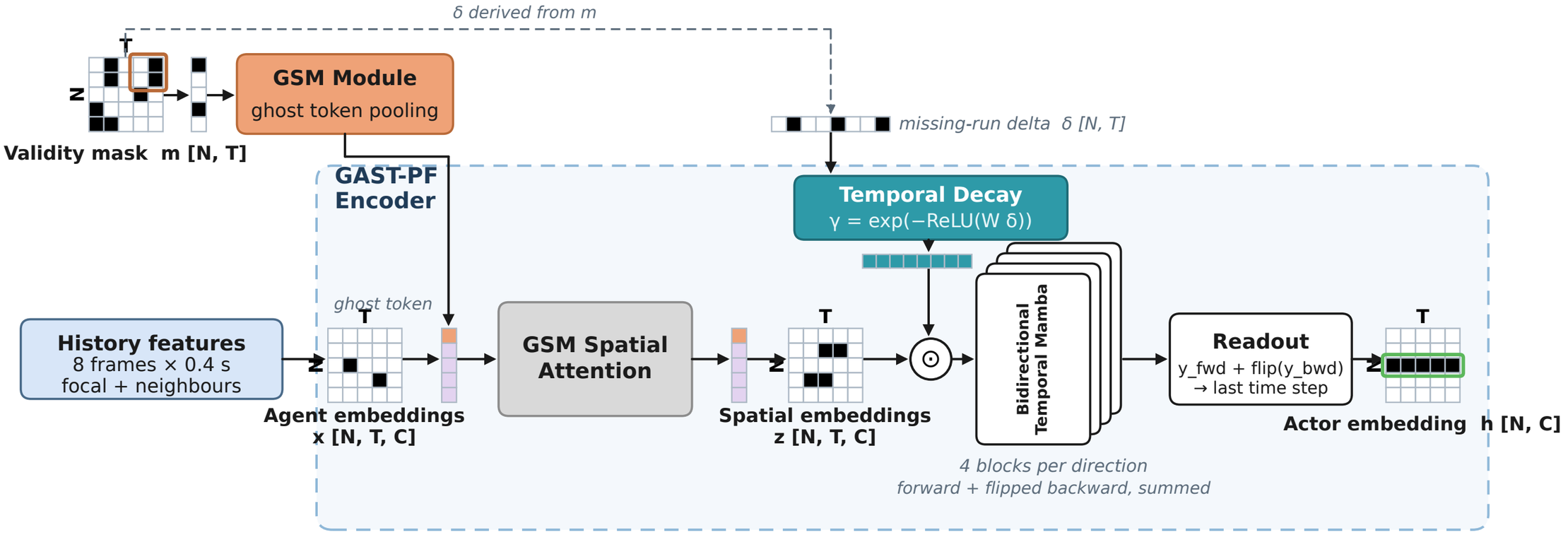
δ derived from m
T
N
GSM Module
ghost token pooling
Validity mask m [N, T]
missing-run delta δ [N, T]
GAST-PF Encoder
Temporal Decay
γ = exp(−ReLU(W δ))
ghost token
History features
8 frames × 0.4 s
focal + neighbours
T
N
Agent embeddings
x [N, T, C]
GSM Spatial Attention
T
N
Spatial embeddings
z [N, T, C]
Bidirectional Temporal Mamba
4 blocks per direction
forward + flipped backward, summed
Readout
y_fwd + flip(y_bwd)
→ last time step
T
N
Actor embedding h [N, C]
N agents
T history
C channels
⊙ element-wise
encoder output h [N, C] → structured future decoder